

# Document

Sovereign AI Workbench • Mangalore Refinery & Petrochemicals Ltd • 2026-09-03 17:32

{"path": "f\_test\_guide.pdf", "title": "F-Test: Mathematical Steps and Applications", "content": "# FTest: Full Mathematical Steps

## 1. Introduction

The Ftest is used to compare two population variances ( $\sigma^2$  and  $\sigma^2$ ). It is based on the ratio of two independent chisquare distributed variables and follows an Fdistribution under the null hypothesis.

## 2. Hypotheses

- **Twotailed test**
  - $(H_0: \sigma_1^2 = \sigma_2^2)$  (the two variances are equal)
  - $(H_a: \sigma_1^2 \neq \sigma_2^2)$  (the variances are not equal)
- **Onetailed tests** (choose direction before data collection)
  - $(H_a: \sigma_1^2 > \sigma_2^2)$  (variance 1 is larger)
  - $(H_a: \sigma_1^2 < \sigma_2^2)$  (variance 1 is smaller)

Significance level:  $(\alpha)$  (commonly 0.05).

## 3. Test Statistic

$$F = \frac{s_1^2}{s_2^2}$$

where

- $(s_1^2)$  and  $(s_2^2)$  are the sample variances.
- By convention, place the **larger variance in the numerator** so that  $(F \geq 1)$ .

### Degrees of freedom

$$\begin{aligned} df_1 &= n_1 - 1 \quad \text{numerator DOF} \\ df_2 &= n_2 - 1 \quad \text{denominator DOF} \end{aligned}$$

The statistic follows an **Fdistribution**:  $(F \sim F(df_1, df_2))$ .

## 4. Decision Rule (CriticalValue Approach)

1. Compute the critical value  $(F_{\alpha, df_1, df_2})$  from the uppertail of the Fdistribution.
2. **Reject  $(H_0)$**  if  $(F > F_{\alpha, df_1, df_2})$  (uppertail).
3. For a **twotailed test** at level  $(\alpha)$ , split the alpha: use  $(\alpha/2)$  for both tails. Reject if

$$F > F_{\alpha/2, df_1, df_2} \quad \text{or} \quad F < F_{1-\alpha/2, df_1, df_2}$$

where the lowertail critical value is the reciprocal of the uppertail value with swapped DOF:

$$F_{1-\alpha/2, df_1, df_2} = \frac{1}{F_{\alpha/2, df_2, df_1}}$$

## 5. pValue Approach

- Compute the cumulative distribution function (CDF) of the Fdistribution:

$$p = 2 \big[ 1 - F_{\text{CDF}}(F) \big] \quad \text{twotailed}$$

- For a onetailed test, use

$$p = 1 - F_{\text{CDF}}(F) \quad \text{upper tail} \quad \text{or} \quad p = F_{\text{CDF}}(F) \quad \text{lower tail}$$

- **Reject  $(H_0)$**  if  $(p < \alpha)$ .

## 6. Worked Example

**Problem:** Compare the variance of batchA ( $n=12, s^2=4.5\text{kg}^2$ ) with batchB ( $n=15, s^2=2.3\text{kg}^2$ ) at  $(\alpha = 0.05)$ .

### 1. Place larger variance in numerator

$$s_1^2 = 4.5, s_2^2 = 2.3 \quad \rightarrow \quad F = \frac{4.5}{2.3} = 1.9565 \approx 1.96$$

### 2. Degrees of freedom

$$df_1 = n_1 - 1 = 11, df_2 = n_2 - 1 = 14$$

### 3. Critical value (upper tail, $\alpha=0.05$ )

From Ftable:  $(F_{0.05, 11, 14} \approx 3.01)$ .

### 4. Decision

Since  $(F = 1.96 < 3.01)$ , we **fail to reject  $(H_0)$** . There is insufficient evidence that the variances differ.

### 5. pvalue (using Python/`scipy.stats.f.cdf`):

$$p = 2 \big[ 1 - F_{\text{CDF}}(1.96) \big] \approx 0.112$$

Since  $(p > 0.05)$ , the conclusion matches the criticalvalue approach.

## 7. Assumptions & Limitations

- **Normality:** Populations must be normally distributed; the Ftest is highly sensitive to nonnormality. Consider Levene's or Bartlett's test as alternatives.
- **Independence:** Samples must be independent.
- **Scale:** Variances are meaningful only for continuous, ratioscale data.

## 8. Python Implementation (scipy.stats)

```
```python
import numpy as np
from scipy import stats
```

### Sample data

```
n1, s1_sq = 12, 4.5 # batch A
n2, s2_sq = 15, 2.3 # batch B
```

### Ensure larger variance in numerator

```
if s1_sq < s2_sq:
    s1_sq, s2_sq = s2_sq, s1_sq
n1, n2 = n2, n1
F = s1_sq / s2_sq
df1, df2 = n1 - 1, n2 - 1
```

### Critical value (upper tail, =0.05)

```
alpha = 0.05
F_crit = stats.f.ppf(1 - alpha, df1, df2)
```

### pvalue (twotailed)

```
p_val = 2 * (1 - stats.f.cdf(F, df1, df2))
print(f"F = {F:.3f} (df={df1}, {df2})")
print(f"Critical F ({alpha}, upper) = {F_crit:.3f}")
print(f"Twotailed pvalue = {p_val:.4f}")
```

### Decision

```
if F > F_crit or p_val < alpha:
    print("Reject H0: variances differ")
else:
    print("Fail to reject H0: variances are equal")
```
```

## 9. References

- Montgomery, D.C., & Runger, G.C. (2011). *Applied Statistics and Probability for Engineers*, 6th ed., Wiley.
- Snedecor, G.W., & Cochran, W.G. (1989). *Statistical Methods*, 8th ed., Iowa State University Press.

*End of document"*

```
}
```